



Technical Service Bulletin

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New Piston Second Ring for QSK60 Tier 2 and Tier 3 Diesel Engines

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK60 CM2150 MCRS
- QSK60 CM850 MCRS

Description of Change

A new second ring is available to be fitted to QSK60 Tier 2 and Tier 3 engines. This ring has a larger end gap that allows pressure to vent quicker from the second land during normal operation.



Figure 1, Second Piston Ring Change

Table 1, Piston Ring End Gap

Part Number	Minimum Value mm [in]	Maximum Value mm [in]
4910253	0.9 [.04]	1.2 [.05]
3645611	1.85 [.07]	2.10 [.08]

Reason for Change

This change has been made as part of a PPS project. Customers have been experiencing liner polishing leading to premature filter plugging and excessive oil consumption. Inspection of the power cylinder shows carbon build up around the piston top land (inspection can be completed via a boroscope).

This second ring change allows pressure to vent quicker from the second land, reducing top ring flutter and oil passing the top ring which leads to carbon formation.

Service Instructions

All engines being re-built should be fitted with the latest ring kit as shown in Table 2. The new part number can be identified by the stamping on the ring.

Service Parts Availability

Service parts are available.

Table 2, Service Parts				
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number
Second Ring	4910253	No	No	3645611
Ring Kit	4955327	No	No	4309263

Part Compatibility

The new ring kit is still compatible with piston, Part Number 3640796.

Production Status

Implemented for production.

Table 3, Production Information		
ESN First	Build Date ¹	Plant
33222453	10 August 2020	Daventry
¹ Engine build date can be found on the engine dataplate.		

Document History

Date	Details
2020-4-24	Module Created
2020-8-25	Made necessary change to Table 2, Production Information.
2021-7-26	Added Table 1.